AJAY KUMAR GARG ENGINEERING COLLEGE, GHAZIABAD

Department of Information Technology

Pre-University Test

Course: B. Tech. **Semester:** IV
Session: 2025-26 **Section:** IT, CSIT, CSE, CS, AIML, CSE (DS), CSE(AIML)

Subject: TAFL **Sub. Code:** BCS-402
Max Marks: 70 **Time:** 3 hrs.

OBE Remarks:

Q.No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
CO No.	1	2	1	1	3	4	5	2	2	2	1	3	3	4	4	5	5
Bloom's Level	L1	L3	L1	L2	L3	L2	L2	L2	L3	L4	L4	L5	L5	L3, L5	L1, L5	L5	L1, L2
Weightage CO3: 16								Weightage CO4: 16							Weightage CO5: 16		

Note: Answer all the sections.

Section-A

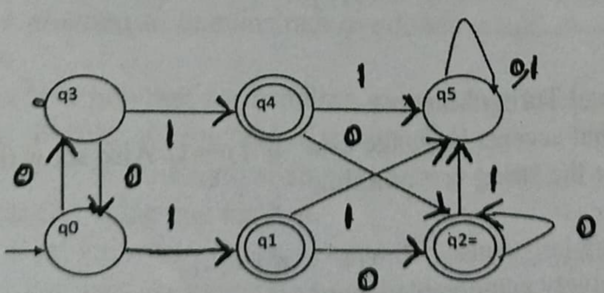
A. Attempt all the parts. (7 X 2 =14)

1. Explain the applications and limitations of finite automata.
2. Design a CFG for the language L= $\{0^m1^n : m \neq n, n >= 1\}$
3. Is every DFA also an NFA? If Yes, then explain.
4. Define and give the difference between positive closure and Kleene closure.
5. Using pumping lemma for Regular languages prove that language $L= 0^{n^2}$ is not regular.
6. Explain briefly about two stack PDA.
7. Explain Multi Tape Turing Machine.

Section-B

B. Attempt Any three. (Q. No. 12 is Compulsory) (3X 7=21)

8. Discuss closure properties (i.e. union, concatenation, complement, intersection and difference) of regular language.
9. Construct a Finite automata (DFA) which accepts all binary numbers whose decimal equivalent is divisible by 4 over $\Sigma = \{0, 1\}$
10. Define NFA. What are various points of difference between NFA and DFA?
11. Minimize the following Automata:



12. Describe various points of difference between Moore & Mealy Machine? Explain the procedure to convert a moore machine into Mealy machine.

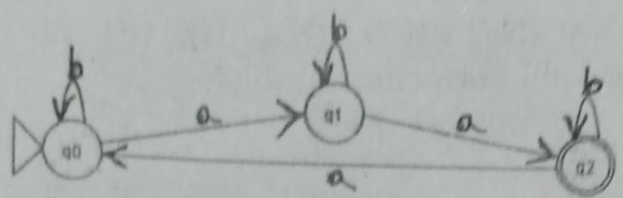
Section-C

C. Attempt all the parts. (5 X 7 = 35)

13. Attempt any one.

a) Find the regular expression of Given FA using Arden's theorem.

Roll No.



b) Let G be the grammar

S → 0B | 1A
A → 0 | 0S | 1AA
B → 1 | 1S | 0BB

Determine the leftmost derivation for the string 00110101

14. Attempt any one.

a) Find the equivalent CFG of the following PDA

P = ({q0, q1,}, {a, b}, {a, z0}, δ, q0, z0) where δ is given by:

δ (q0, a, z0) = (q0, az0)
δ (q0, a, a) = (q1, aa)
δ (q1, a, a) = (q1, ε)
δ (q1, ε, z0) = (q1, ε)

b) Construct PDA equivalent to the following CFG G = ({S, A}, {0,1}, P, S} where P is defined by:

S → 0S1 | A
A → 1A0 | S | ε

15.Attempt any one.

a) Convert the following PDA into its equivalent CFG. Transition function is defined as:

$$\delta(q_0, 0, Z_0) = \{(q_0, 0Z_0)\}$$
$$\delta(q_0, 0, 0) = \{(q_0, 00)\}$$
$$\delta(q_0, 1, 0) = \{(q_1, \in)\}$$
$$\delta(q_1, 1, 0) = \{(q_1, \in)\}$$
$$\delta(q_1, \in, Z_0) = \{(q_2, \in)\}$$

b) Construct a PDA to accept the $L = \{ WcW^R \mid W \in (a, b)^+ \}$

16. Attempt any one.

a) Write Short notes on Universal Turing Machine.

b) Construct Turing Machine that accepts language $L = \{a^{2n} b^n \mid n >= 1\}$. Also show the Instantaneous description for the string w = aaaabb.

17. Attempt any one.

a) Design a TM for the following language: $L = \{ a^n b^n c^n \mid n \geq 1\}$

b) Define Recursive and recursively enumerable language

Faculty Sign

HoD Sign